WELLSLING · CONCEPT OPTIONS

WellSling concept options

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish chose option A and graft option (a) on 2026-10-25 (WSL-DEC-001 D15, D16)
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade). The design carried since kit 1.7 is option A; options B and C added with different working principles; all three modelled at D3 and scored against WSL-URS-001

WellSling concept options

Recommendation: option A, the tripod and hand winch carried since kit 1.7, scores 224 of 265 against 198 for option B (portable gantry and two chain hoists) and 177 for option C (sheerlegs and draught haul). B is 11.6 % behind A, just outside the 10 % band, so one option was recommended, and Amish chose it on 2026-10-25 (WSL-DEC-001 D15, WSL-DDR-003). Method: STANDARDS section 21.

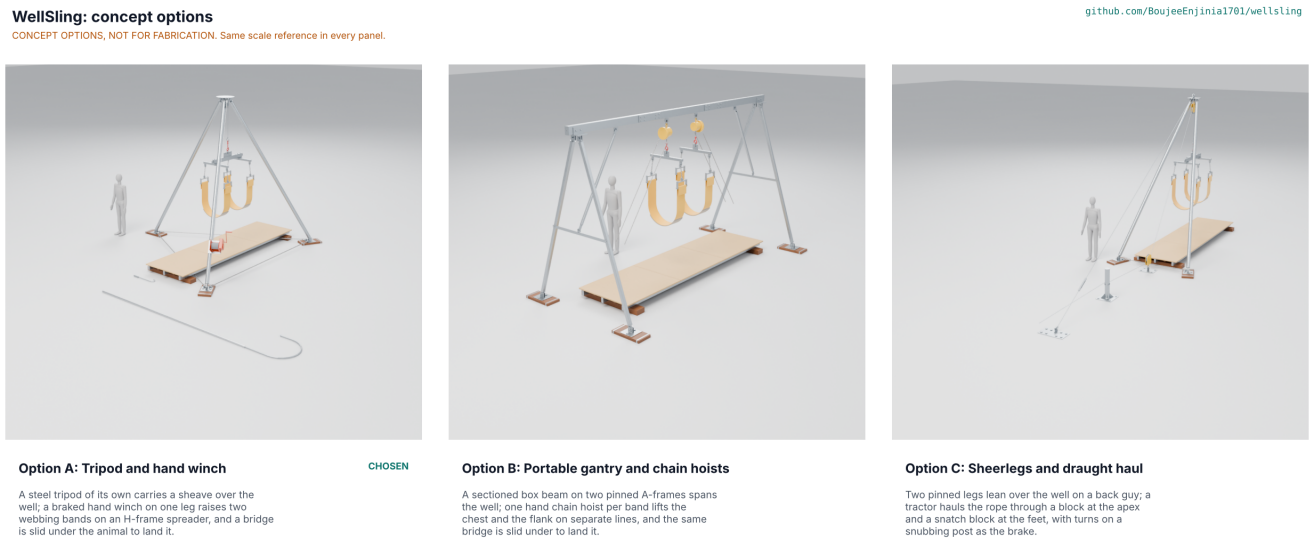


Figure 1. The three options at one scale, rendered from the D3 models in `cad/src/model.py` *(the Cycles picture is made by the coordinator with* `.kit/options.py` *).*

> **Safety:** Every option lifts a 600 kg animal over an open well and is worked from the rim; none is used before a proof-load test at 1.5 times the hook load. Option C puts a tractor on the rope and relies on a person holding the brake turns, which is its main safety weakness.

The options

Option A: Tripod and hand winch

A steel tripod of its own carries a sheave over the well; a braked hand winch on one leg raises two webbing bands on an H-frame spreader, and a bridge is slid under the animal to land it. Three 88.9 mm legs pinned to a head plate stand on staked foot plates and timber pads, tied by spread chains; the 8 mm rope runs from a 1,000 kg spur-gear winch with an automatic load-pressure brake up leg 1 and over a 200 mm sheave. Each band's two eyes hang on band end yokes, adjusting chains and shackles from the cross bars of the spreader. First-order numbers: hook load 662 kg, rope factor 6.0, winch at 87 % on its third layer, crank at most 200 N, about 30 minutes for a 15 m lift, tips only at 19.4 deg, heaviest piece 32.0 kg (WSL-CAL-001). Key risks: animal tolerance of the bands, passer push force, setup time at its limit. From a co-design partner it needs real wells, a test pit and a veterinarian.

Option B: Portable gantry and chain hoists

A sectioned box beam on two pinned A-frames spans the well; one hand chain hoist per band lifts the chest and the flank on separate lines, and the same bridge is slid under to land it. The 5.8 m beam (RHS 140 x 80 x 4, three bolted sections) sits in saddles on A-frames of SHS 60 x 3 legs 2.7 m either side of the well, with knee braces to the beam. Each 1 t hoist hangs on a shackle under the beam and lifts a band bar (a cross bar with a top lug) carrying the same chains, yokes and bands as A. First-order numbers: beam factor 3.1, legs 39 on Euler, hand chain pull about 120 N per hoist (assumed), heaviest beam section 25.7 kg (after the fix D14), about 20 to 30 minutes with two operators. Its best feature: chest and flank are lifted separately, so the animal can be kept level. Key risks: thin beam margin, the beam lifted onto saddles at 3.3 m, two operators pulling hand chains over the well mouth, 15 m of load chain per hoist.

Option C: Sheerlegs and draught haul

Two pinned legs lean over the well on a back guy; a tractor hauls the rope through a block at the apex and a snatch block at the feet, with turns on a snubbing post as the brake. The legs (CHS 88.9 x 3.2, 4.74 m) stand on staked feet 2.4 m from the well axis, tied by a chain; a 10 mm guy with a turnbuckle runs to a staked anchor plate 7 m back. The rope runs from the hook over the top block, down to a snatch block on its own staked anchor, three turns round a staked post and on to the tractor; the spreader, bands and bridge are A's. First-order numbers: guy tension 6.5 kN, leg factor 7.6 on Euler, about 400 N to hold the load on three turns, the lift itself about a minute at a tractor's creep speed. Key risks: the guy anchor's holding in soft ground, a jerk or missed stop from the tractor, no automatic brake, three anchors to set.

First-order numbers

Table 1. Options compared. Parts and mass from `cad/fidelity.json` (every modelled part, including fasteners and welds; catalogue items such as the winch and rope drawn in part); costs are first-order estimates from the option A BOM rates, with the chain hoist price cited in WSL-MKT-001.

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	3,093 (priced BOM)	About 3,400 (estimate, plus or minus 30 %)	About 3,000 (estimate, plus or minus 30 %) plus a tractor
Mass of the modelled kit (kg)	522.0	654.4	552.8
Parts in the D3 model	351 (86 fasteners, 100 welds)	427 (164 fasteners, 86 welds)	312 (67 fasteners, 81 welds)
Output or throughput	15 m lift in about 30 min	About 20 to 30 min (estimate)	About 1 min of haul once rigged
Energy or effort	152 kJ at the crank, up to 200 N, crew of three in turns	About 160 kJ at two hand chains (estimate), about 120 N each	Tractor; about 400 N on the brake turns
Crew	3 to 4	4 (two at the hand chains)	4 and a tractor driver
STEM check	Within known science; 8 rows open for TRL 4	Within known science; B3 plausible	Within known science; C1 and C4 plausible

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5).

Criterion	Source	Weight	A	B	C	Reason
Nobody goes down the well	UR-001	3	5	5	5	All three are worked from the rim; the bands are passed by the passer in every option
Lifts a 600 kg animal	UR-002	3	5	4	4	A: every part at a factor of 3 or more; B: beam at 3.1, the least margin; C: frame fine, guy anchor holding unknown (C1)
Wells 15 m deep	UR-003	3	5	4	5	A and C: rope length is cheap; B: two hoists each need 15 m of load chain and a long hand chain loop
Wells 3 m across, weak rims	UR-004	3	4	4	4	All keep pads 0.7 m or more back from a 3 m rim
Ready in 20 minutes	UR-005	3	3	2	2	A: 20 min estimate, at the limit; B: two A-frames, three beam sections and 24 bolts lifted to 3.3 m; C: three staked anchors, a guy and a tractor to bring up
Short briefing	UR-006	3	4	3	2	A: one crank, one brake; B: two hoists must be pulled together to keep the animal level; C: driver, signaller and brake hand must work as one
Stops and holds	UR-007	3	5	5	2	A and B: automatic load brakes; C: only the turns on the post, held by a person
Gentle on the animal	UR-008	3	4	5	3	B can lift chest and flank separately to keep the animal level; C lifts fast and can jerk
Lands without swinging	UR-009	3	5	5	4	Same slid-in bridge in all three; C's hook is off the leg plane and swings more if the tractor surges
Human power only	UR-010	3	4	4	2	A: 200 N at the crank; B: hand chain about 120 N per hoist (assumed); C needs a tractor
Frame does not tip	UR-011	3	5	4	3	A: tips only at 19.4 deg; B: knee braces hold the beam line; C: stands on its guy and one anchor
Nobody falls in	UR-012	3	4	3	4	B's hand chains hang over the well mouth and are pulled from the rim edge
Two-person pieces	UR-013	2	5	3	4	A: 32.0 kg at most; B: 25.7 kg beam sections, but the beam is assembled and lifted to 3.3 m; C: legs about 31 kg
Affordable to share	UR-014	2	3	2	3	First-order cost A USD 3,093, B about USD 3,400, C about USD 3,000 plus a tractor
Lives outdoors	UR-015	2	4	4	4	All steel galvanised; timber pads and deck
Local repair	UR-016	2	4	3	4	A: one bought winch; B: two bought chain hoists; C: blocks and a post a workshop can service
Free to copy	UR-017	1	5	5	5	No insert in the bands and a separate passer in every option
		2	2	1	2	All over the USD 1,500 target; B the most

Criterion	Source	Weight	A	B	C	Reason
Cost against the value-engineering target	STANDARDS s21					
Buildable with local skills	STANDARDS s21	2	4	4	4	Plate, tube, box section and timber in all three
Safety	STANDARDS s21	3	4	4	2	C: a tractor on the line, a person holding the brake turns and an anchor of unknown holding
Maintenance	STANDARDS s21	1	4	3	4	Hoist chains need care in mud; A's winch and C's blocks are simple
Weighted total (of 265)			**224**	**198**	**177**	

Recommendation

Option A is recommended: it meets every high-priority user need on paper, holds the load on an automatic brake, needs nothing but people at the well, and has the most margin on the frame. Option B is the runner-up at 198 (11.6 % behind, outside the 10 % band) and its best idea, lifting the chest and the flank separately to keep the animal level, can be grafted onto A without changing its working principle: replace one band's two adjusting chains with a 0.75 t lever hoist so its height can be changed under load (WSL-DEC-001 D16; on 2026-10-25 Amish chose to keep the chains for now and let the TRL 4 dummy trial show whether the hoist is needed). Option C is the fastest lift but scores lowest on control and safety. Amish decided on 2026-10-25 ("i approve of your recommendations"): option A, recorded in the decisions register and WSL-DDR-003; option A stays `CHOSEN` in the model.